Grid Risk Prediction: Interpretation, Uncertainty, and Limitations

SYNTHETIC EDUCATIONAL DOCUMENT — NOT AN OFFICIAL SOURCE

Purpose: Provide background material for retrieval and chatbot testing in the SmartGrid AI prototype.

This synthetic document is designed to help explain predictive analytics in an electricity advisory application. It does not validate any particular SmartGrid AI model.

1. Prediction is not certainty

A model estimates an outcome based on patterns in its training data. A predicted probability is not a guarantee that an event will or will not occur.

2. Synthetic data

Synthetic data can be useful for prototyping, testing software, and demonstrating machine-learning workflows. It may not represent real-world distributions, rare events, regional differences, or changing operating conditions.

3. Data quality

Missing values, inconsistent timestamps, measurement errors, geographic mismatches, and changes in definitions can affect model quality. Data provenance and preprocessing should be documented.

4. Leakage and validation

Features that would not be available at prediction time must not be used. Time-based problems should use chronological validation. Hyperparameter tuning should not use the final test set.

5. Classification thresholds

A threshold converts a probability into a class. Changing the threshold changes false positives and false negatives. The threshold should be justified by the cost of each type of error and validated against representative data.

6. Regression uncertainty

A single predicted demand value hides uncertainty. Prediction intervals or scenario ranges may provide a more realistic picture when supported by the model and validation data.

7. Human oversight

Operational decisions should be made by qualified grid operators and utilities using validated tools and real-time system information. A prototype advisory tool should not issue control instructions or claim official status.

8. Responsible presentation

Display model version, data period, geographic scope, key limitations, and whether the output is based on synthetic or real observations.

End of synthetic document. Verify operational facts with current official sources before real-world use.